Automobile Engineering • Semester II

Basic Electrical and Electronics Engineering

Course Code: 2051 | Credits: 4 | Revision: 2026 | Theory

Fundamental principles of DC/AC circuits, electrical machines, automotive power sources, and electronic automation.

Download Study Notes (PDF)

Official Source Declaration

This lesson is strictly based on the **SITTTR Kerala Revision 2026 Syllabus** for Course Code **2051** (Basic Electrical and Electronics Engineering). All modules, learning outcomes, instructional hours, and assessment patterns are extracted directly from the official PDF.

- **Program:** Automobile Engineering
- **Course Type:** Theory (L:4, T:0, P:0, S:0)
- **Total Hours:** 60 Instructional + 90 Self-learning

Course Dashboard

Teaching Scheme

Component	Hours/Week
Lecture (L)	4
Self-Learning (SS)	6

Total Semester Hours	60 (Instructional)

Prerequisites

- Basic Science - Physics (2002A)
- Basic Mathematics (1002)

How to Study this Lesson

This course is designed for Automobile Engineering students. Focus on how electrical principles apply to vehicle systems like charging, ignition, and ECUs.

Study Tip: Use the interactive Ohm's Law calculator in Module 1 and the Transformer EMF calculator in Module 2 to practice numerical problems.

Course Outcomes (CO)

CO No.	Outcome Description	Bloom's Level
CO1	Explain fundamental principles of DC/AC circuits, electromagnetic induction, and 3-phase systems to solve basic problems.	Understand
CO2	Describe construction, working principles, and applications of DC motors, AC induction motors, and transformers.	Understand
CO3	Discuss working of automotive power sources (batteries, generators, alternators) and industrial heating.	Understand
CO4	Explain electronic switching devices (MOSFET, SCR, Transistors), logic gates, and automotive automation architecture.	Understand

Syllabus Coverage

Module	Title	Instructional Hours	Weightage
I	Electrical Circuits	15	25%

Module	Title	Instructional Hours	Weightage
II	Electric Motors and Transformers	15	25%
III	Electrical Sources and Power Utilisation	15	25%
IV	Fundamentals of Electronics and Automation	15	25%

Learning Roadmap

Step 1: Fundamentals (Module I)

Master Voltage, Current, and Ohm's Law. Understand how AC and 3-phase power works.

Step 2: Conversion (Module II)

Learn how electrical energy is converted to mechanical energy (Motors) and how voltage levels are changed (Transformers).

Step 3: Generation & Storage (Module III)

Study how cars generate power (Alternators) and store it (Batteries).

Step 4: Control (Module IV)

Understand the electronic "brain" of the vehicle using transistors, logic gates, and sensors.

Module I: Electrical Circuits

15 Hours

1.1 Voltage, Current, Resistance, Power and Energy

These are the building blocks of any electrical system. In an automobile, the battery provides the **Voltage** (Pressure), the wiring carries the **Current** (Flow), and the components like bulbs or motors provide **Resistance**.

- **Voltage (V):** Electric potential difference. Unit: Volt (V).
- **Current (I):** Rate of flow of charge. Unit: Ampere (A).
- **Resistance (R):** Opposition to current flow. Unit: Ohm (Ω).
- **Power (P):** Rate of doing work. $P = V \times I$. Unit: Watt (W).
- **Energy (E):** Power consumed over time. $E = P \times t$. Unit: Joule (J) or kWh.

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> Voltage [illegible]. Current [illegible]. Resistance [illegible]. [illegible] 12V [illegible], [illegible].

Ohm's Law

At constant temperature, the current through a conductor is directly proportional to the voltage across it. $V = I \times R$.

Ohm's Law Calculator

Voltage (V): Enter Volts

Resistance (Ω): Enter Ohms

Result: -- A

1.2 Electromagnetic Induction & AC Circuits

Faraday's Law: Whenever a conductor cuts magnetic flux, an EMF is induced in it. This is the principle behind vehicle alternators.

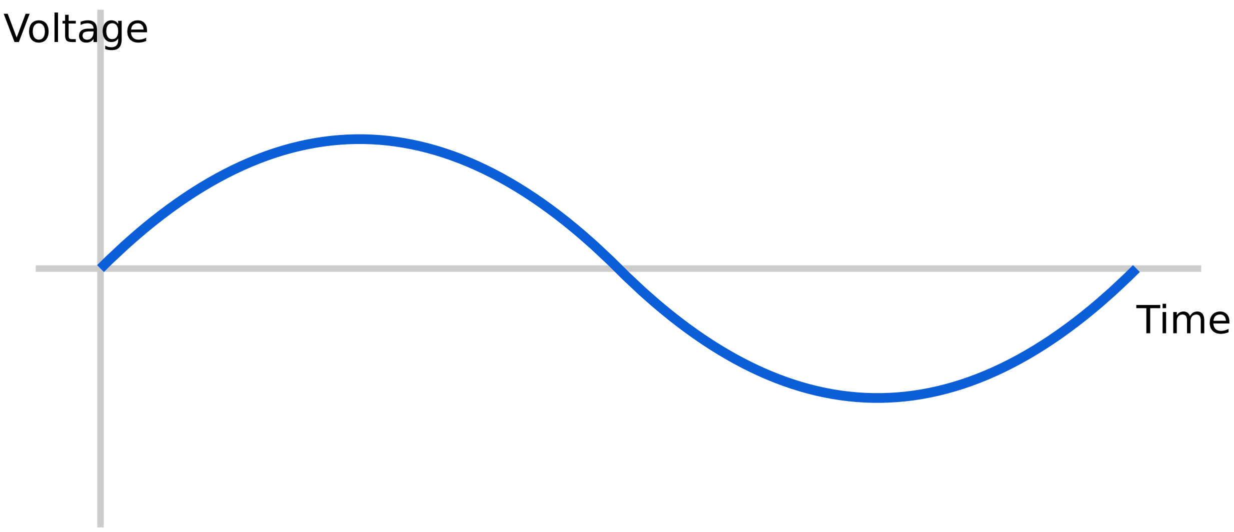

Fig 1.1: Sinusoidal AC Waveform

Key AC Terms:

- **RMS Value:** The effective value of AC. $V_{rms} = V_{max} / \sqrt{2}$.
- **Frequency (f):** Number of cycles per second. Unit: Hertz (Hz).
- **Power Factor:** Ratio of real power to apparent power.

Module II: Electric Motors and Transformers

15 Hours

2.1 DC Motors

A DC motor converts DC electrical energy into mechanical energy. It works on the principle that a current-carrying conductor placed in a magnetic field experiences a force.

Types of DC Motors in Automobiles

Type	Application
Series Motor	Starter Motor (High starting torque)
Shunt Motor	Wiper motors, cooling fans

Compound Motor	Heavy industrial drives

2.2 Transformers

A static device that transfers electrical energy between circuits through electromagnetic induction, usually changing voltage levels.

EMF Equation:

$$E = 4.44 \times f \times \Phi m \times N$$

f = frequency, Φm = max flux, N = number of turns

Turns Ratio (K): $V_2/V_1 = N_2/N_1 = I_1/I_2$.

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Module III: Electrical Sources and Power Utilisation

15 Hours

3.1 DC Generator & Alternator

In modern cars, the **Alternator** is used to charge the battery and power electrical systems while the engine is running. It produces AC, which is then rectified to DC.

3.2 Batteries

Automobiles primarily use **Lead-Acid Batteries**. They consist of lead plates in an electrolyte of sulfuric acid.

- **Lead-Acid:** Reliable, high cold-cranking amps.
- **Alkaline (Nickel-Iron):** Robust, long life, but lower efficiency.
- **VRLA:** Valve Regulated Lead Acid (Maintenance-free).

3.3 Electric Heating

Joule's Law: $H = I^2Rt$. Heat produced is proportional to the square of current, resistance, and time.

- **Induction Heating:** Used for melting metals (Eddy currents).
- **Dielectric Heating:** Used for non-conductors (plastics, textiles).

Module IV: Fundamentals of Electronics and Automation

15 Hours

4.1 Diodes and Rectifiers

A **PN Junction Diode** allows current in one direction only (Forward Bias). **Rectifiers** convert AC to DC.

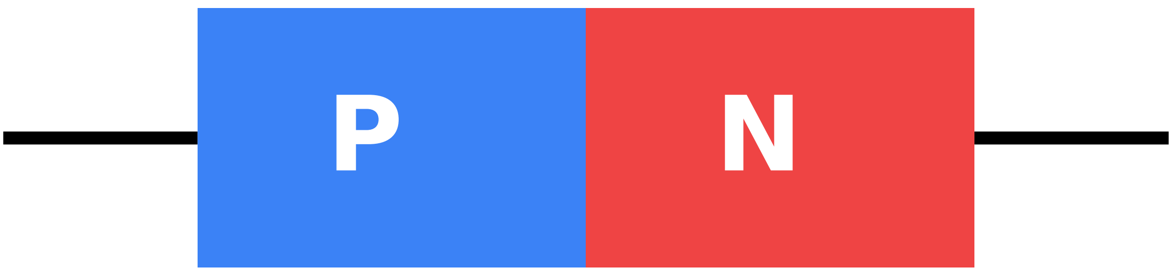

Fig 4.1: PN Junction Diode Structure

4.2 Transistors and MOSFETs

Transistors act as **switches** or **amplifiers**. In automotive ECUs, Power MOSFETs are used for high-speed switching of fuel injectors and ignition coils.

4.3 Logic Gates & Automation

Logic gates (AND, OR, NOT, NAND, NOR) form the basis of digital control. **Universal Gates:** NAND and NOR.

Automotive Control System: Sensors → ECU → Actuators.

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> [illegible] (Sensors) [illegible] [illegible], ECU [illegible] [illegible] [illegible], [illegible] (Actuators) [illegible]. [illegible], [illegible] [illegible] Thermistor [illegible].

Suggested Student Activities

No	Activity Description
1	Identify battery, fuse box, and wiring harness in a vehicle.
4	Calculate total resistance and power of headlamp circuits (Series/Parallel).
9	Disassemble a used lead-acid battery cell safely and sketch components.
13	Decode resistor color bands and verify with a multimeter.

Formula Bank

Ohm's Law

$$V = I \times R$$

Electrical Power

$$P = V \times I$$

Transformer Ratio

$$N1/N2 = V1/V2$$

Joule's Heating

$$H = I^2Rt$$

AC RMS Voltage

$$Vrms = Vmax / 1.414$$

Induction Motor Slip

$$s = (Ns - Nr) / Ns$$

Visual Library

Circuit Symbols

Standard symbols for Resistors, Capacitors, Diodes, and Transistors used in automotive schematics.

Motor Construction

Internal views of DC Shunt Motors and Squirrel Cage Induction Motors.

Assessment Methodology

Assessment	Marks	Details
CIE (Internal)	40	Activities (15), Series Exams (20), Attendance (5)

ESE (Semester)	60	2.5 Hour Written Exam

ESE Question Paper Pattern

- **Part A:** 8 questions (2 per module) × 1 mark = 8 Marks
- **Part B:** 8 questions (2 out of 3 per module) × 3 marks = 24 Marks
- **Part C:** 4 questions (1 set per module with choice) × 7 marks = 28 Marks

Module-wise Questions

Module I

State Ohm's Law and its limitations.

Ohm's Law states that the current flowing through a conductor is directly proportional to the potential difference across its ends, provided temperature and other physical conditions remain constant ($V=IR$). Limitations: It does not apply to non-ohmic conductors (like semiconductors) or when temperature varies.

Module II

Why is a DC series motor used as a starter motor in cars?

A DC series motor provides very high starting torque, which is necessary to overcome the inertia and compression of the internal combustion engine during startup.

Module III

Explain the purpose of an alternator in a vehicle.

The alternator converts mechanical energy from the engine into AC electrical energy, which is rectified to DC to charge the battery and power the vehicle's electrical components (lights, ECU, etc.) while the engine is running.

Module IV

What are universal gates? Why are they called so?

NAND and NOR gates are called universal gates because any other logic gate (AND, OR, NOT) can be constructed using only these gates.

Practice Model Paper

Note: This is a practice paper based on the official pattern, not an official examination paper.

Part A (1 Mark each)

1. Define RMS value of AC.
2. What is the unit of electrical energy?
3. Define slip in an induction motor.
4. Mention one application of a transformer.

Part B (3 Marks each)

1. Compare single-phase and three-phase systems.
2. Explain the working principle of a DC motor.
3. Differentiate between lead-acid and alkaline batteries.
4. Draw the V-I characteristics of a PN junction diode.

Quick Revision

Key Points

- $V=IR$, $P=VI$, $E=Pt$.
- Transformers change voltage, not frequency.
- Alternators charge the battery in cars.
- Diodes = Rectifiers; Transistors = Switches.

Exam Checklist

- [ ] Definitions of V, I, R, P, E.
- [ ] Faraday's Laws of Induction.

- [ ] DC Motor types and applications.
- [ ] Transformer EMF equation.
- [ ] Logic gate truth tables.

Glossary

ECU
Electronic Control Unit - The "computer" in a vehicle.

Rectifier
A circuit that converts AC to DC.

Stator
The stationary part of an electric motor/generator.

Rotor
The rotating part of an electric motor/generator.

Semiconductor
Material with conductivity between a conductor and insulator (e.g., Silicon).

References

- Fundamentals of Electrical Engineering and Electronics - B.L. Theraja (S. Chand)
- Electronic Devices and Circuits - G.K. Mittal (Khanna Publishers)
- Automobile Electrical and Electronic Systems - Tom Denton (Routledge)
- NPTEL: Basic Electrical Technology